

AN ELEMENTARY INTRODUCTION TO MICROECONOMICS



KUTUSH KUMAR, KUTUM PAKHI

The business meaning matters as much as the arithmetic.

This section builds the idea of decisions on the balance sheet before moving to application.

Balance sheet. A statement listing what a business owns and what it owes on one date..

The term balance sheet gives the chapter a precise name for this part of the problem.

Asset. Something a business owns that has value, such as cash, stock or a van..

The term asset gives the chapter a precise name for this part of the problem.

Liability. Money a business owes to others, such as a bank loan or an unpaid supplier bill..

The term liability gives the chapter a precise name for this part of the problem.

Equity. What is left for the owners when liabilities are subtracted from assets..

The term equity gives the chapter a precise name for this part of the problem.

The central question in decisions on the balance sheet is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what a financial manager does, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

2.3 DECISIONS ON THE BALANCE SHEET

The purpose here is to make decisions on the balance sheet concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Balance sheet** means A statement listing what a business owns and what it owes on one date. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Asset** means Something a business owns that has value, such as cash, stock or a van. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Liability** means Money a business owes to others, such as a bank loan or an unpaid supplier bill. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Equity** means What is left for the owners when liabilities are subtracted from assets. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For decisions on the balance sheet, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a

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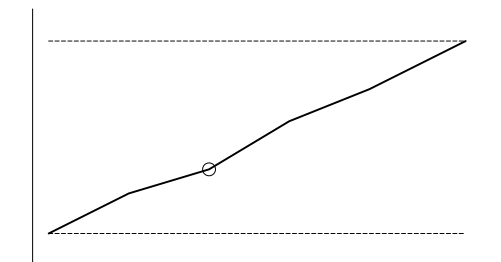


FIG. 2.1. The Financial Manager's Decisions figure 1

2.1 THE THREE BIG DECISIONS

This section builds the idea of the three big decisions before moving to application.

Financial manager. The person who decides how a business finds, uses and protects its money.

The term financial manager gives the chapter a precise name for this part of the problem.

Investment decision. A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money..

The term investment decision gives the chapter a precise name for this part of the problem.

Financing decision. A choice about where the money comes from: owners' own funds, a loan, or new owners..

The term financing decision gives the chapter a precise name for this part of the problem.

Dividend. A share of a company's profit paid out to its owners instead of being kept in the business..

The term dividend gives the chapter a precise name for this part of the problem.

The central question in the three big decisions is what changes when the underlying decision changes.

The purpose here is to make the three big decisions concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Financial manager** means The person who decides how a business finds, uses and protects its money. In a small shop this may be the owner. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Investment decision** means A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financing decision** means A choice about where the money comes from: owners' own funds, a loan, or new owners. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Dividend** means A share of a company's profit paid out to its owners instead of being kept in the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches An Elementary Introduction to Microeconomics one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

REVIEW QUESTIONS

01.T1 say in plain words what finance covers and how it differs from accounting and from management

01.T2 tell profit from cash flow using a new set of sales and collection figures

01.T3 compare sole proprietorship, partnership and company for a new tailoring shop

PAUSE AND CHECK 01.P1 After the finance definition, state the rule in your own words.

PAUSE AND CHECK 01.P2 After profit versus cash flow, state the rule in your own words.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.

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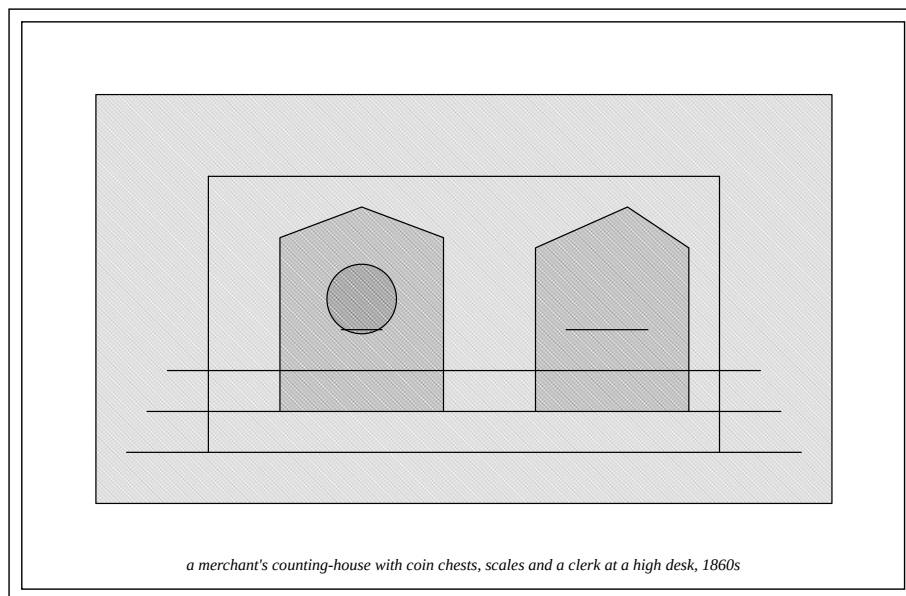
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CHAPTER I

What Finance Is

A practical question about what finance is



This chapter introduces what finance is for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A calculation can be correct and still answer the wrong question. For finance in plain words, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

1.2 FINANCE BESIDE ITS NEIGHBOURING SUBJECTS

The purpose here is to make finance beside its neighbouring subjects concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating efficiency** means How much useful output a business gets from the money and effort it uses. A shop that sells more per taka of stock is more efficient. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sole proprietorship** means A business owned and run by one person, who keeps all profit and answers for all debts. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Partnership** means A business owned by two or more people who agree to run it together and share its profit. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For finance beside its neighbouring subjects, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

The business meaning matters as much as the arithmetic.

This section builds the idea of finance beside its neighbouring subjects before moving to application.

Operating efficiency. How much useful output a business gets from the money and effort it uses.

The term operating efficiency gives the chapter a precise name for this part of the problem.

Sole proprietorship. A business owned and run by one person, who keeps all profit and answers for all debts..

The term sole proprietorship gives the chapter a precise name for this part of the problem.

Partnership. A business owned by two or more people who agree to run it together and share its profit..

The term partnership gives the chapter a precise name for this part of the problem.

The central question in finance beside its neighbouring subjects is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

1.3 THE FINANCE FUNCTION INSIDE A FIRM

The purpose here is to make the finance function inside a firm concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Company** means A business set up as a legal person of its own, owned through shares, so the owners are separate from the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the finance function inside a firm, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to what finance is. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

This section builds the idea of the finance function inside a firm before moving to application.

Company. A business set up as a legal person of its own, owned through shares, so the owners are separate from the business..

The term company gives the chapter a precise name for this part of the problem.

The central question in the finance function inside a firm is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

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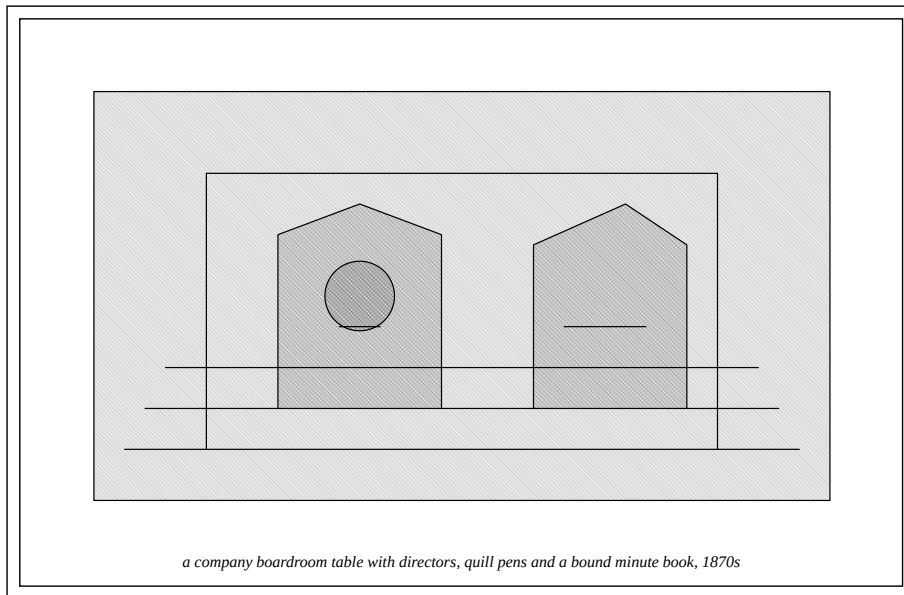
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CHAPTER III

The Goal of the Firm and Agency Theory

A practical question about the goal of the firm and agency theory



This chapter introduces the goal of the firm and agency theory for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For profit maximisation and its limits, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

3.2 WEALTH MAXIMISATION

The purpose here is to make wealth maximisation concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Stakeholder** means Anyone affected by a firm's actions, such as staff, customers, lenders and the community. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Principal** means A person who hires another to act for them. A shop owner who hires a manager is a principal. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Agent** means A person who acts for a principal and makes decisions for the principal. A hired shop manager is an agent. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Agency problem** means A conflict that arises when an agent's own interests differ from those of the principal. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of wealth maximisation before moving to application.

Stakeholder. Anyone affected by a firm's actions, such as staff, customers, lenders and the community..

The term stakeholder gives the chapter a precise name for this part of the problem.

Principal. A person who hires another to act for them.

The term principal gives the chapter a precise name for this part of the problem.

Agent. A person who acts for a principal and makes decisions for the principal.

The term agent gives the chapter a precise name for this part of the problem.

Agency problem. A conflict that arises when an agent's own interests differ from those of the principal..

The term agency problem gives the chapter a precise name for this part of the problem.

The central question in wealth maximisation is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

* * *

5.3 COMPARING STREAMS OF CASH FLOWS

The purpose here is to make comparing streams of cash flows concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Cash outflow** means Money going out of a business or from an investor, shown as a negative number. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Mixed stream** means A series of cash flows of different sizes in different periods. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

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A calculation can be correct and still answer the wrong question. For comparing streams of cash flows, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to present value and discounting. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ $(1 + \text{rate})$. Steps: Tk 100,000 $\times$ $(1 + 0.10)$ = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

The second pass checks both the arithmetic and the financial interpretation.

units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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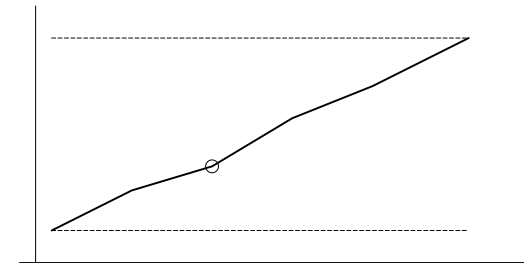


FIG. 5.1. Present Value and Discounting figure 1

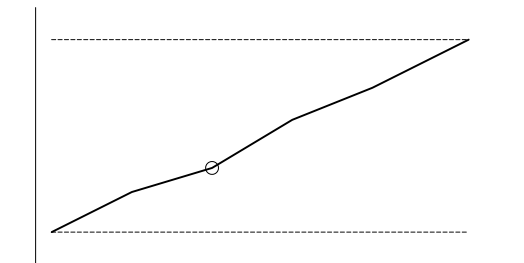


FIG. 5.2. Present Value and Discounting figure 2

5.1 DISCOUNTING A SINGLE SUM

The purpose here is to make discounting a single sum concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Present value** means What a sum due in the future is worth today. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discounting** means Finding present value by removing the interest a sum could earn before it arrives. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of discounting a single sum before moving to application.

Present value. What a sum due in the future is worth today..

The term present value gives the chapter a precise name for this part of the problem.

Discounting. Finding present value by removing the interest a sum could earn before it arrives..

The term discounting gives the chapter a precise name for this part of the problem.

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SUMMARY

The central purpose of The Goal of the Firm and Agency Theory is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Berle and Means, The Modern Corporation and Private Property (1932), and Jensen and Meckling (1976) on agency costs. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Richard Price, the 1772 Appeal on the National Debt, and the British sinking fund of 1786 that his compound-interest argument helped to inspire. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

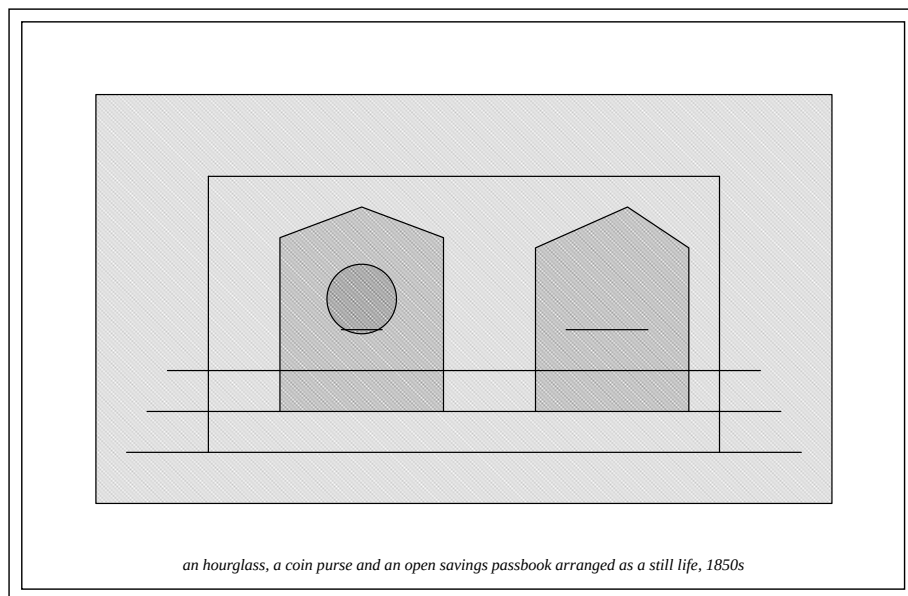
04.T1 explain why a taka today beats a taka next year, with new figures

04.T2 compute simple and compound interest on Tk 50,000 at 12% for four years

04.T3 use a factor table to find a future value for a new rate and period

PAUSE AND CHECK 04.P1 After opportunity cost, state the rule in your own words.

PAUSE AND CHECK 04.P2 After compound interest, state the rule in your own words.



CHAPTER IV

Interest and Future Value

A practical question about interest and future value



This chapter introduces interest and future value for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A short restatement keeps the financial meaning attached to the calculation.

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In this chapter, **Interest** means The payment a borrower makes to a lender for the use of money, or the reward a saver earns. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Interest rate** means The interest for one period written as a percentage of the sum, for example 8% a year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For why time changes the value of money, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

4.2 SIMPLE AND COMPOUND INTEREST

The purpose here is to make simple and compound interest concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Principal sum** means The original amount lent, borrowed or deposited, before any interest is added. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Simple interest** means Interest calculated only on the principal sum, so it is the same every period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Interest. The payment a borrower makes to a lender for the use of money, or the reward a saver earns..

The term interest gives the chapter a precise name for this part of the problem.

Interest rate. The interest for one period written as a percentage of the sum, for example 8% a year..

The term interest rate gives the chapter a precise name for this part of the problem.

The central question in why time changes the value of money is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of simple and compound interest before moving to application.

Principal sum. The original amount lent, borrowed or deposited, before any interest is added..

The term principal sum gives the chapter a precise name for this part of the problem.

Simple interest. Interest calculated only on the principal sum, so it is the same every period..

The term simple interest gives the chapter a precise name for this part of the problem.

Compound interest. Interest calculated on the principal sum and also on interest already earned..

The term compound interest gives the chapter a precise name for this part of the problem.

Future value. What a sum today will be worth at a later date when it earns interest..

The term future value gives the chapter a precise name for this part of the problem.

The central question in simple and compound interest is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of future value of a single sum before moving to application.

Compounding period. The length of time after which interest is added to the balance, such as a year or six months..

The term compounding period gives the chapter a precise name for this part of the problem.

Timeline. A line marking each period, with the cash flow written at the point where it occurs..

The term timeline gives the chapter a precise name for this part of the problem.

In this chapter, **Compound interest** means Interest calculated on the principal sum and also on interest already earned. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Future value** means What a sum today will be worth at a later date when it earns interest. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For simple and compound interest, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

4.3 FUTURE VALUE OF A SINGLE SUM

The purpose here is to make future value of a single sum concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Compounding period** means The length of time after which interest is added to the balance, such as a year or six months. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Timeline** means A line marking each period, with the cash flow written at the point where it occurs. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

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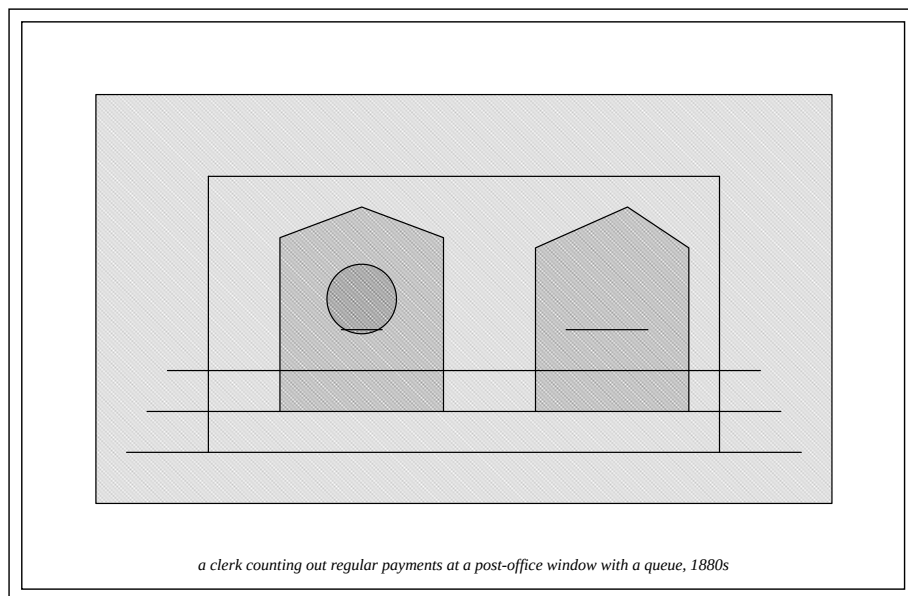
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CHAPTER VI

Annuities

A practical question about annuities



This chapter introduces annuities for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what an annuity is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

6.2 FUTURE VALUE OF AN ANNUITY

The purpose here is to make future value of an annuity concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Ordinary annuity** means An annuity whose payments fall at the end of each period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For future value of an annuity, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

6.3 PRESENT VALUE OF AN ANNUITY

The purpose here is to make present value of an annuity concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of future value of an annuity before moving to application.

Ordinary annuity. An annuity whose payments fall at the end of each period..

The term ordinary annuity gives the chapter a precise name for this part of the problem.

The central question in future value of an annuity is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of present value of an annuity before moving to application.

* * *

8.3 OTHER MEASURES OF A PROJECT

The purpose here is to make other measures of a project concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Profitability index** means Present value of future cash flows divided by the initial investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For other measures of a project, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

8.4 CHOOSING AMONG METHODS

The purpose here is to make choosing among methods concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

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A calculation can be correct and still answer the wrong question. For choosing among methods, keep the decision in view. Ask what the business is trying

This section builds the idea of other measures of a project before moving to application.

Profitability index. Present value of future cash flows divided by the initial investment..

The term profitability index gives the chapter a precise name for this part of the problem.

The central question in other measures of a project is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of choosing among methods before moving to application.

The central question in choosing among methods is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to annuities. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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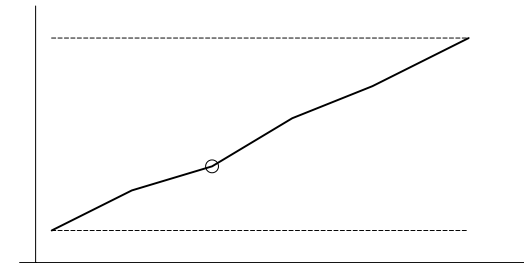


FIG. 8.1. Evaluating Projects: Payback and Net Present Value figure 1

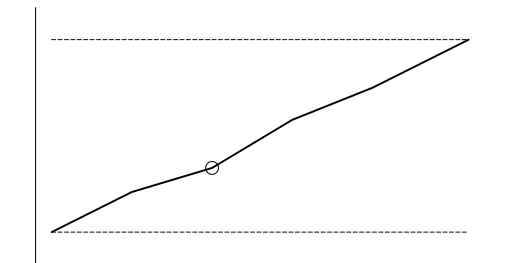


FIG. 8.2. Evaluating Projects: Payback and Net Present Value figure 2

8.1 PAYBACK PERIOD

The purpose here is to make payback period concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Payback period** means The time a project takes to return its initial investment from its cash flows. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discounted payback period** means The payback time when each cash flow is first discounted to present value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of payback period before moving to application.

Payback period. The time a project takes to return its initial investment from its cash flows..

The term payback period gives the chapter a precise name for this part of the problem.

Discounted payback period. The payback time when each cash flow is first discounted to present value..

The term discounted payback period gives the chapter a precise name for this part of the problem.

each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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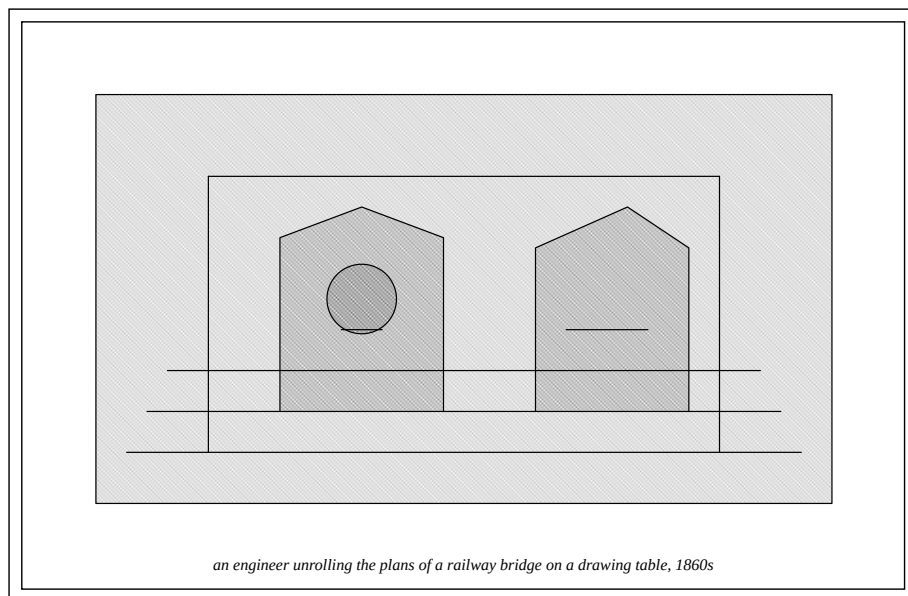
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A short restatement keeps the financial meaning attached to the calculation.

07.T4 find the operating cash flow from a new set of sales, costs, depreciation and tax

PAUSE AND CHECK 07.P1 After the steps, state the rule in your own words.

PAUSE AND CHECK 07.P2 After relevant cash flows, state the rule in your own words.



CHAPTER VII

Capital Budgeting: Process and Cash Flows

A practical question about capital budgeting: process and cash flows



This chapter introduces capital budgeting: process and cash flows for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and

The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For why firms budget for capital, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

7.2 THE STEPS OF THE PROCESS

The purpose here is to make the steps of the process concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Relevant cash flow** means A cash flow that changes because of the decision being studied. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sunk cost** means Money already spent that cannot be recovered whatever you decide now. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Initial investment** means The cash flows at the start of a project to buy and prepare the asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Operating cash flow** means The cash a project brings in each year after paying its cash costs and tax. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the steps of the process before moving to application.

Relevant cash flow. A cash flow that changes because of the decision being studied..

The term relevant cash flow gives the chapter a precise name for this part of the problem.

Sunk cost. Money already spent that cannot be recovered whatever you decide now..

The term sunk cost gives the chapter a precise name for this part of the problem.

Initial investment. The cash flows at the start of a project to buy and prepare the asset..

The term initial investment gives the chapter a precise name for this part of the problem.

Operating cash flow. The cash a project brings in each year after paying its cash costs and tax..

The term operating cash flow gives the chapter a precise name for this part of the problem.

The central question in the steps of the process is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of cash flows from an investment before moving to application.

Terminal cash flow. The cash received or paid when a project ends, such as the sale of scrap..

The term terminal cash flow gives the chapter a precise name for this part of the problem.

Depreciation. The share of an asset's cost charged against income each year as the asset wears out..

The term depreciation gives the chapter a precise name for this part of the problem.

The central question in cash flows from an investment is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the steps of the process, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

7.3 CASH FLOWS FROM AN INVESTMENT

The purpose here is to make cash flows from an investment concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Terminal cash flow** means The cash received or paid when a project ends, such as the sale of scrap. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Depreciation** means The share of an asset's cost charged against income each year as the asset wears out. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For cash flows from an investment, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to capital budgeting: process and cash flows. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the capital asset pricing model before moving to application.

Unsystematic risk. Risk that affects one firm or industry only; diversification can remove it..

The term unsystematic risk gives the chapter a precise name for this part of the problem.

Beta. A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market..

The term beta gives the chapter a precise name for this part of the problem.

Market return. The average return of all investments in a market during a period..

The term market return gives the chapter a precise name for this part of the problem.

The central question in the capital asset pricing model is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For diversification and beta, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

11.3 THE CAPITAL ASSET PRICING MODEL

The purpose here is to make the capital asset pricing model concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Unsystematic risk** means Risk that affects one firm or industry only; diversification can remove it. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Beta** means A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Market return** means The average return of all investments in a market during a period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the capital asset pricing model, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a

CHAPTER IX

Risk and Return

A practical question about risk and return



This chapter introduces risk and return for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

In this chapter, **Risk** means The chance that the actual result will differ from the result you expect. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Business risk** means Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For return and risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

9.2 DESCRIBING UNCERTAINTY

The purpose here is to make describing uncertainty concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Financial risk** means Extra risk that comes from borrowing, because interest must be paid whatever the sales. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Probability distribution** means A list of possible results with the chance of each, the chances adding to one. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Risk. The chance that the actual result will differ from the result you expect..

The term risk gives the chapter a precise name for this part of the problem.

Business risk. Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing..

The term business risk gives the chapter a precise name for this part of the problem.

The central question in return and risk is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of describing uncertainty before moving to application.

Financial risk. Extra risk that comes from borrowing, because interest must be paid whatever the sales..

The term financial risk gives the chapter a precise name for this part of the problem.

Probability distribution. A list of possible results with the chance of each, the chances adding to one..

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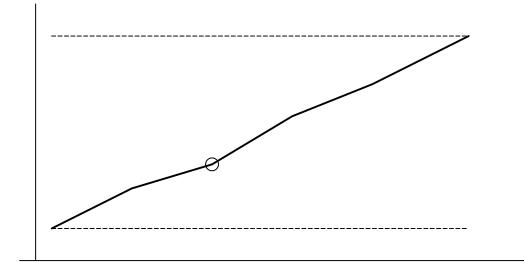


FIG. 11.1. Required Return, CAPM and Market Efficiency figure 1

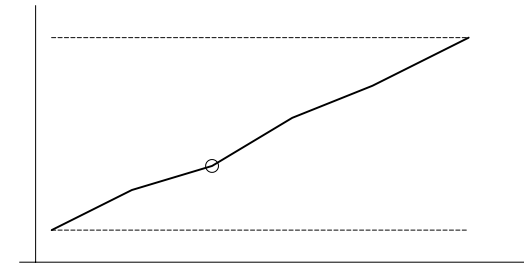


FIG. 11.2. Required Return, CAPM and Market Efficiency figure 2

11.1 RISK PREMIUM AND REQUIRED RETURN

The purpose here is to make risk premium and required return concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Required return** means The least return an investor will accept for taking a given risk. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Risk-free rate** means The return on an investment with no risk of loss, such as a short-term government security. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of risk premium and required return before moving to application.

Required return. The least return an investor will accept for taking a given risk..

The term required return gives the chapter a precise name for this part of the problem.

Risk-free rate. The return on an investment with no risk of loss, such as a short-term government security..

The term risk-free rate gives the chapter a precise name for this part of the problem.

The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For measuring risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to risk and return. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

IN HISTORY

A historical case shows how the chapter's financial problem appeared in practice.

Long-Term Capital Management, whose heavy borrowing led to a rescue organised by the Federal Reserve Bank of New York on 23 September 1998. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

10.T1 say what leverage means and tell its three kinds apart with new data

10.T2 find the break-even point and degrees of operating, financial and total leverage for a new firm

10.T3 explain in plain words why high leverage raises risk

PAUSE AND CHECK 10.P1 After break-even point, state the rule in your own words.

PAUSE AND CHECK 10.P2 After degree of operating leverage, state the rule in your own words.

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Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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PAUSE AND CHECK 09.P1 After probability distribution, state the rule in your own words.

PAUSE AND CHECK 09.P2 After standard deviation, state the rule in your own words.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of financial leverage before moving to application.

Operating leverage. The effect of fixed operating costs on how EBIT responds to sales.

The term operating leverage gives the chapter a precise name for this part of the problem.

Financial leverage. The effect of fixed financing payments on how owners' earnings respond to EBIT.

The term financial leverage gives the chapter a precise name for this part of the problem.

Total leverage. The combined effect of operating and financial leverage on how owners' earnings respond to sales.

The term total leverage gives the chapter a precise name for this part of the problem.

The central question in financial leverage is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For operating leverage, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

10.3 FINANCIAL LEVERAGE

The purpose here is to make financial leverage concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating leverage** means The effect of fixed operating costs on how EBIT responds to sales. Measured by the degree of operating leverage (DOL), the percentage change in EBIT divided by the percentage change in sales. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

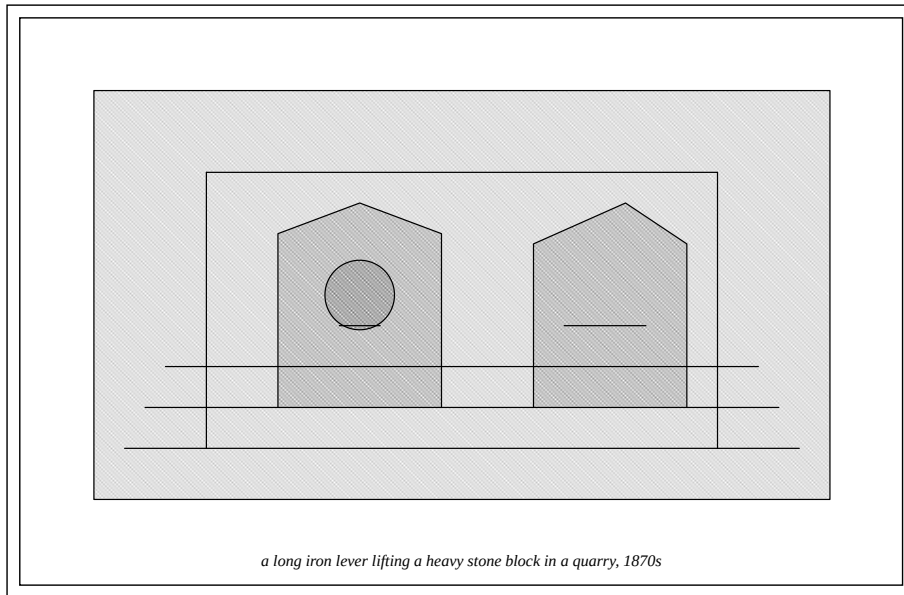
In this chapter, **Financial leverage** means The effect of fixed financing payments on how owners' earnings respond to EBIT. Measured by the degree of financial leverage (DFL). The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Total leverage** means The combined effect of operating and financial leverage on how owners' earnings respond to sales. Measured by the degree of total leverage (DTL), which equals DOL times DFL. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For financial leverage, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a



CHAPTER X

Leverage

A practical question about leverage



This chapter introduces leverage for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

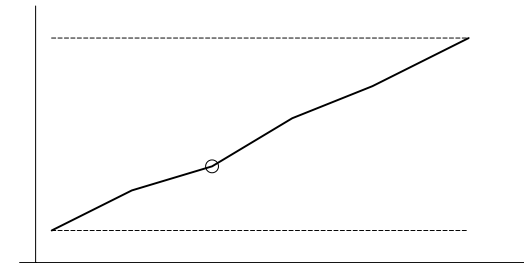


FIG. 10.1. Leverage figure 1

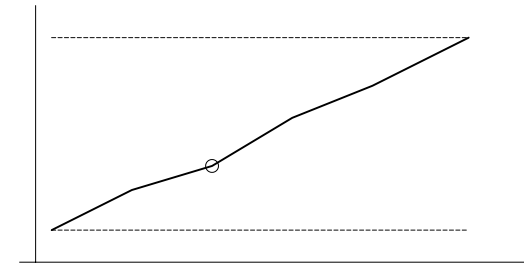


FIG. 10.2. Leverage figure 2

10.1 COSTS AND THE BREAK-EVEN POINT

The purpose here is to make costs and the break-even point concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Leverage** means The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Fixed cost** means A cost that stays the same however many units are sold, such as shop rent. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of costs and the break-even point before moving to application.

Leverage. The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings..

The term leverage gives the chapter a precise name for this part of the problem.

Fixed cost. A cost that stays the same however many units are sold, such as shop rent..

The term fixed cost gives the chapter a precise name for this part of the problem.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Required Return, CAPM and Market Efficiency is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs

A short restatement keeps the financial meaning attached to the calculation.

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The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

Long-term financing. Money raised for more than about five years, or with no fixed repayment date..

The term long-term financing gives the chapter a precise name for this part of the problem.

Trade credit. Credit from a supplier who lets a buyer pay some days after delivery..

The term trade credit gives the chapter a precise name for this part of the problem.

Cash discount. A price cut a supplier gives for paying early, such as 2% off if paid within 10 days..

The term cash discount gives the chapter a precise name for this part of the problem.

Inventory. The goods a business holds for sale, such as the books on a shop's shelves..

The term inventory gives the chapter a precise name for this part of the problem.

Accounts receivable. Money that customers owe a business for goods already delivered..

The term accounts receivable gives the chapter a precise name for this part of the problem.

The central question in sources of short-term finance is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of cycles, float and estimating needs before moving to application.

In this chapter, **Long-term financing** means Money raised for more than about five years, or with no fixed repayment date. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Trade credit** means Credit from a supplier who lets a buyer pay some days after delivery. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

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In this chapter, **Accounts receivable** means Money that customers owe a business for goods already delivered. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

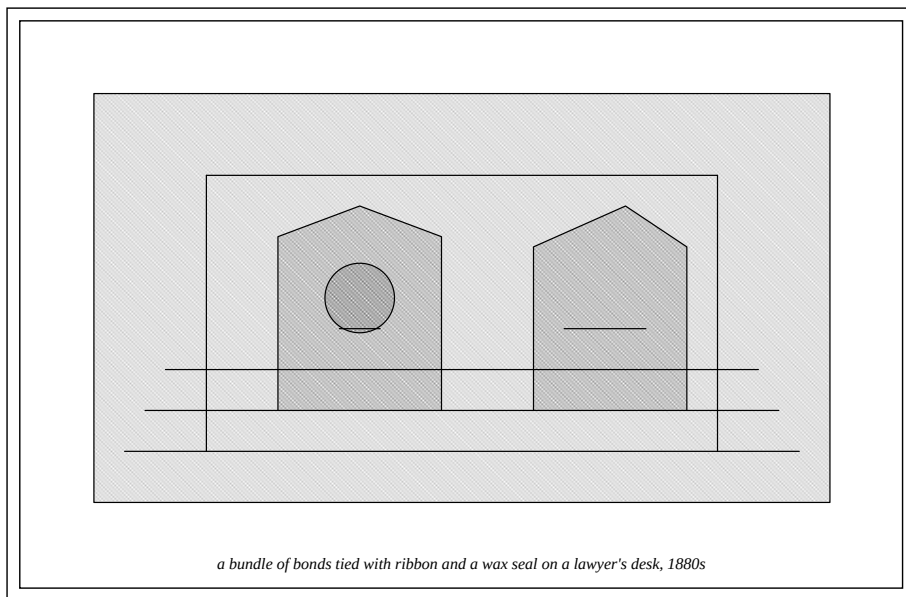
Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For sources of short-term finance, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

14.3 CYCLES, FLOAT AND ESTIMATING NEEDS

The purpose here is to make cycles, float and estimating needs concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the



CHAPTER XII

Valuation of Bonds

A practical question about valuation of bonds



This chapter introduces valuation of bonds for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

This section builds the idea of working capital and how to finance it before moving to application.

Working capital. The money a business has tied up in everyday trading: cash, stock and money owed to it..

The term working capital gives the chapter a precise name for this part of the problem.

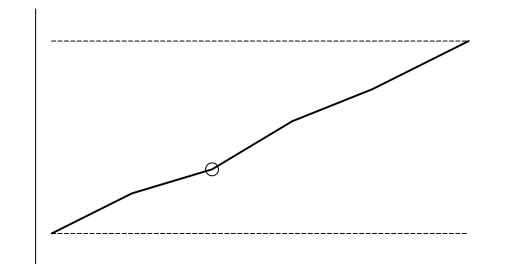


FIG. 14.1. *Working Capital and Short-Term Financing figure 1*

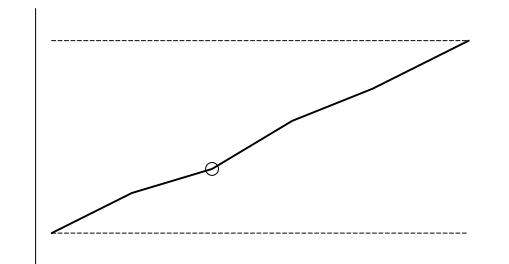


FIG. 14.2. *Working Capital and Short-Term Financing figure 2*

14.1 WORKING CAPITAL AND HOW TO FINANCE IT

The purpose here is to make working capital and how to finance it concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Working capital** means The money a business has tied up in everyday trading: cash, stock and money owed to it. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Bond** means A loan made to a company or government, repaid on a set date, with interest paid on the way. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Face value** means The amount a bond repays at its end, for example Tk 1,000. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the idea of valuation, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

12.2 WHAT A BOND IS

The purpose here is to make what a bond is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Coupon rate** means The yearly interest of a bond, as a percentage of its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Maturity** means The date on which a bond or loan must be repaid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Yield to maturity** means The yearly return an investor earns by buying a bond at today's price and holding it to maturity. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Bond. A loan made to a company or government, repaid on a set date, with interest paid on the way..

The term bond gives the chapter a precise name for this part of the problem.

Face value. The amount a bond repays at its end, for example Tk 1,000..

The term face value gives the chapter a precise name for this part of the problem.

The central question in the idea of valuation is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of what a bond is before moving to application.

Coupon rate. The yearly interest of a bond, as a percentage of its face value..

The term coupon rate gives the chapter a precise name for this part of the problem.

Maturity. The date on which a bond or loan must be repaid..

The term maturity gives the chapter a precise name for this part of the problem.

Yield to maturity. The yearly return an investor earns by buying a bond at today's price and holding it to maturity..

The term yield to maturity gives the chapter a precise name for this part of the problem.

13.T2 value a preferred stock and a growing common stock from new data

13.T3 explain why growth raises value but also needs a required return above growth

PAUSE AND CHECK 13.P1 After common stock, state the rule in your own words.

PAUSE AND CHECK 13.P2 After perpetuity, state the rule in your own words.

finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

12.4 YIELD AND PRICE

The purpose here is to make yield and price concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For yield and price, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to valuation of bonds. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

This section builds the idea of yield and price before moving to application.

The central question in yield and price is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

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each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Valuation of Bonds is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

A short restatement keeps the financial meaning attached to the calculation.

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The chapter has connected its main terms, method, and business decision.

Constant growth model. The dividend valuation model for dividends that grow at the same rate every year..

The term constant growth model gives the chapter a precise name for this part of the problem.

The central question in valuing common stock is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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In this chapter, **Constant growth model** means The dividend valuation model for dividends that grow at the same rate every year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For valuing common stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

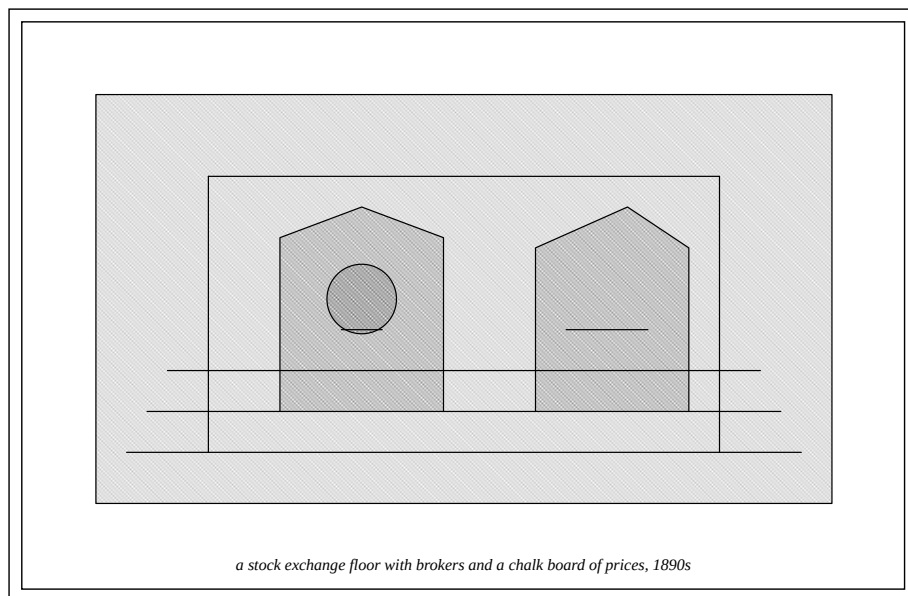
Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to valuation of shares. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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CHAPTER XIII

Valuation of Shares

A practical question about valuation of shares



This chapter introduces valuation of shares for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

This section builds the idea of common stock before moving to application.

Common stock. Shares that give owners a vote and a claim on profit after all other claims are met..

The term common stock gives the chapter a precise name for this part of the problem.

Residual claim. The right to whatever is left after lenders and preferred owners have been paid..

The term residual claim gives the chapter a precise name for this part of the problem.

The central question in common stock is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

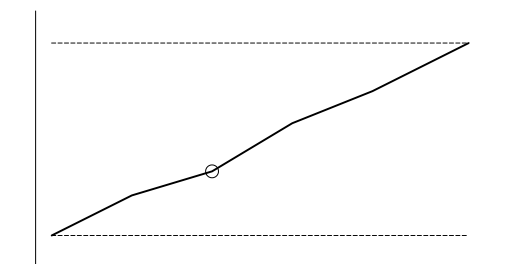


FIG. 13.1. Valuation of Shares figure 1

13.1 COMMON STOCK

The purpose here is to make common stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Common stock** means Shares that give owners a vote and a claim on profit after all other claims are met. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Residual claim** means The right to whatever is left after lenders and preferred owners have been paid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For common stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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A.1 A1.1 PERCENTAGES AND PERCENTAGE CHANGE

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

A.2 A1.2 RATIOS AND AVERAGES

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

A.3 A1.3 REARRANGING AN EQUATION AND ROUNDING

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

A.4 A1.4 READING A GRAPH

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

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PAUSE AND CHECK 14.P1 After matching principle, state the rule in your own words.

PAUSE AND CHECK 14.P2 After cash conversion cycle, state the rule in your own words.

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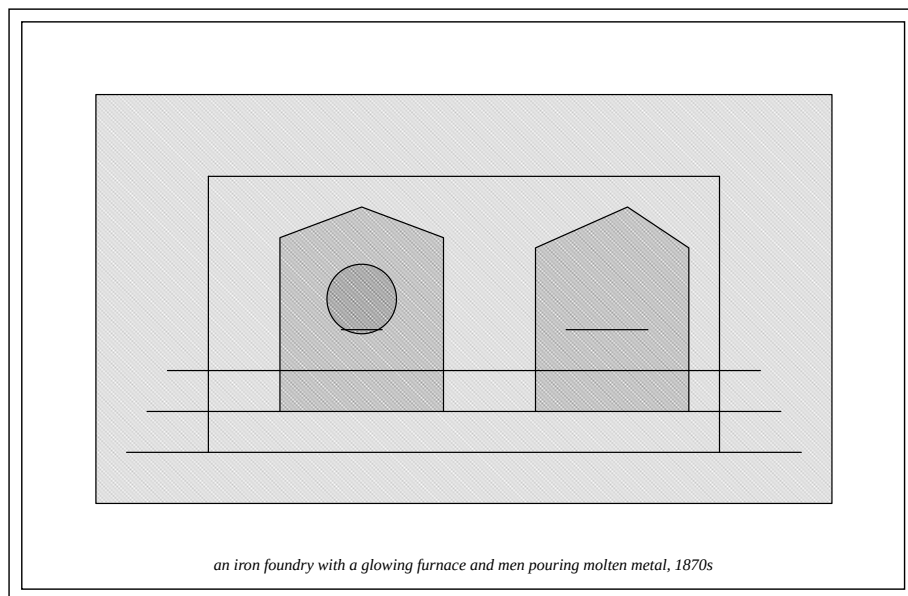
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SUMMARY

The central purpose of Leasing is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify



CHAPTER XV

Long-Term Financing and the Capital Market

A practical question about long-term financing and the capital market



This chapter introduces long-term financing and the capital market for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A calculation can be correct and still answer the wrong question. For term loans and the features of financing, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

15.2 DEBT AND EQUITY AS SOURCES

The purpose here is to make debt and equity as sources concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Debenture** means A bond backed only by the issuer's general promise to pay, not by a particular asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Retained earnings** means Profit kept in the business instead of being paid out as dividends. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Capital market** means The market in which long-term money is raised by selling bonds and shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For debt and equity as sources, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

The business meaning matters as much as the arithmetic.

This section builds the idea of debt and equity as sources before moving to application.

Debenture. A bond backed only by the issuer's general promise to pay, not by a particular asset..

The term debenture gives the chapter a precise name for this part of the problem.

Retained earnings. Profit kept in the business instead of being paid out as dividends..

The term retained earnings gives the chapter a precise name for this part of the problem.

Capital market. The market in which long-term money is raised by selling bonds and shares..

The term capital market gives the chapter a precise name for this part of the problem.

The central question in debt and equity as sources is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Tax shield** means The tax saved because a cost, such as interest or depreciation, reduces taxable income. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **After-tax cash outflow** means A payment's cost after subtracting the tax it saves. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the lease-or-buy decision, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to leasing. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

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The central question in the lease-or-buy decision is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

In this chapter, **Cost of capital** means The return a firm must earn on its investments to satisfy those who supply its money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Weighted average cost of capital** means The average cost of a firm's different sources of money, each weighted by its share of the total. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

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A calculation can be correct and still answer the wrong question. For the cost of capital, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to long-term financing and the capital market. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

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The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

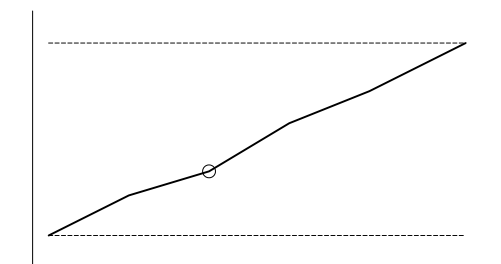


FIG. 16.1. Leasing figure 1

16.1 WHAT LEASING IS

The purpose here is to make what leasing is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Lease** means A contract in which the owner of an asset lets another party use it for regular payments. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Lessor** means The owner who lets the asset out under a lease. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Lessee** means The user who pays to use the asset under a lease. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what leasing is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a

This section builds the idea of what leasing is before moving to application.

Lease. A contract in which the owner of an asset lets another party use it for regular payments..

The term lease gives the chapter a precise name for this part of the problem.

Lessor. The owner who lets the asset out under a lease..

The term lessor gives the chapter a precise name for this part of the problem.

Lessee. The user who pays to use the asset under a lease..

The term lessee gives the chapter a precise name for this part of the problem.

The central question in what leasing is is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Long-Term Financing and the Capital Market is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs

A short restatement keeps the financial meaning attached to the calculation.

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The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Dhaka Stock Exchange: its share-price boom of 2010 and the sharp fall of early 2011. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

15.T1 sort given facts into short, intermediate and long-term financing

15.T2 describe the main sources of long-term money for a new company with their strengths and limits

15.T3 explain what retained earnings cost the firm

15.T4 compute the weighted average cost of capital for new figures

PAUSE AND CHECK 15.P1 After the features of financing, state the rule in your own words.

PAUSE AND CHECK 15.P2 After capital-market issues, state the rule in your own words.

GLOSSARY

Accounts payable. Money a business owes its suppliers for goods already received. (*Chapter XIV*)

Accounts receivable. Money that customers owe a business for goods already delivered. (*Chapter XIV*)

After-tax cash outflow. A payment's cost after subtracting the tax it saves. (*Chapter XVI*)

Agency cost. A cost the principal bears to reduce the agency problem or from the loss it still causes. (*Chapter III*)

Agency problem. A conflict that arises when an agent's own interests differ from those of the principal. (*Chapter III*)

Agent. A person who acts for a principal and makes decisions for the principal. A hired shop manager is an agent. (*Chapter III*)

Annuity. A series of equal payments made at equal intervals for a fixed number of periods. (*Chapter VI*)

Annuity due. An annuity whose payments fall at the start of each period. (*Chapter VI*)

Asset. Something a business owns that has value, such as cash, stock or a van. (*Chapter II*)

Balance sheet. A statement listing what a business owns and what it owes on one date. (*Chapter II*)

Beta. A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market. (*Chapter XI*)

Bond. A loan made to a company or government, repaid on a set date, with interest paid on the way. (*Chapter XII*)

Break-even point. The sales volume at which total revenue equals total cost, so profit is zero. (*Chapter X*)

Business risk. Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing. (*Chapter IX*)

Capital asset pricing model. A formula for required return: risk-free rate plus beta times the market risk premium. (*Chapter XI*)

Capital budgeting. Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring. (*Chapter VII*)

Capital expenditure. Spending on an asset that will help a business for more than one year. (*Chapter VII*)

Capital gain. The profit from selling an asset for more than its purchase price. (*Chapter XIII*)

Capital market. The market in which long-term money is raised by selling bonds and shares. (*Chapter XV*)

Cash conversion cycle. The operating cycle minus the days a business takes to pay its suppliers. (*Chapter XIV*)

Cash discount. A price cut a supplier gives for paying early, such as 2% off if paid within 10 days. (*Chapter XIV*)

Cash flow. Money that actually moves into or out of a business. Paying a supplier Tk 5,000 is a cash flow of minus Tk 5,000. (*Chapter I*)

Dividend valuation model. A way of valuing a share as the present value of the dividends it is expected to pay. (*Chapter XIII*)

Earnings per share. Earnings left for owners of common stock divided by the number of shares. (*Chapter X*)

EBIT. Earnings before interest and tax: sales minus operating costs. (*Chapter X*)

Efficient market. A market in which prices already reflect the available information. (*Chapter XI*)

Equity. What is left for the owners when liabilities are subtracted from assets. (*Chapter II*)

Expected value. The average of possible results, each weighted by its probability. (*Chapter IX*)

Face value. The amount a bond repays at its end, for example Tk 1,000. (*Chapter XII*)

Factor table. A table of ready-made multipliers that saves repeating a long calculation. (*Chapter IV*)

Finance. The activity of finding money, using it, and judging whether the result was worth it. A shop owner who borrows to buy stock does finance. (*Chapter I*)

Financial lease. A long lease, not cancellable, that covers most of the asset's life and works like a loan. (*Chapter XVI*)

Financial leverage. The effect of fixed financing payments on how owners' earnings respond to EBIT. Measured by the degree of financial leverage (DFL). (*Chapter X*)

Financial manager. The person who decides how a business finds, uses and protects its money. In a small shop this may be the owner. (*Chapter II*)

Financial planning. Estimating the money a business will need and receive in future, and arranging how to meet the gap. (*Chapter II*)

Financial risk. Extra risk that comes from borrowing, because interest must be paid whatever the sales. (*Chapter IX*)

Financing decision. A choice about where the money comes from: owners' own funds, a loan, or new owners. (*Chapter II*)

Fixed cost. A cost that stays the same however many units are sold, such as shop rent. (*Chapter X*)

Float. Money tied up in payments that are sent but not yet usable, such as a cheque in the post. (*Chapter XIV*)

Future value. What a sum today will be worth at a later date when it earns interest. (*Chapter IV*)

Incentive. A reward, such as a bonus, that encourages a person to act in a chosen way. (*Chapter III*)

Independent project. A project that can be accepted or rejected without affecting any other project. (*Chapter VII*)

Initial investment. The cash flows at the start of a project to buy and prepare the asset. (*Chapter VII*)

Initial public offering. A company's first sale of shares to the public. (*Chapter XV*)

Instalment. One of a series of regular payments used to repay a loan or build a sum. (*Chapter VI*)

R

Rate of return, 59, 64, 66

Relevant cash flow, 51, 56

Required return, 4, 13, 21, 29, 37, 45, 53, 61, 69, 80, ...

Residual claim, 100

Retained earnings, 119, 124

Return, 4, 13, 15, 21, 29, 35, 37, 45, 53, 58–59, ...

Rights issue, 120

Risk, 4, 7, 13, 15, 18, 21, 23, 29, 32, 37, ...

Risk premium, 84–85, 87, 90

Risk-free rate, 84, 87

S

Sale-and-leaseback, 127

Security market line, 87

Semiannual interest, 94

Share price, 18, 105

Shareholder, 18, 120

Short-term financing, 108–109, 111, 114

Simple interest, 27

Sole proprietorship, 3, 8

Stakeholder, 19

Standard deviation, 68, 72–73, 90

Sunk cost, 51

Systematic risk, 85

T

Tax shield, 128, 131

Term loan, 118–119

Terminal cash flow, 52

Time value of money, 26

Timeline, 28

Total leverage, 78–79, 82

Trade credit, 110

Treasurer, 11, 15

U

Underwriter, 120

Unsystematic risk, 86, 90

V

Valuation, 7, 15, 23, 32, 40, 48, 55, 64, 72, 82, ...

Variable cost, 77

Variance, 68, 72

percentage change in EBIT divided by the percentage change in sales. (*Chapter X*)

Opportunity cost. What you give up by choosing one use of money instead of the best alternative. (*Chapter IV*)

Ordinary annuity. An annuity whose payments fall at the end of each period. (*Chapter VI*)

Partnership. A business owned by two or more people who agree to run it together and share its profit. (*Chapter I*)

Payback period. The time a project takes to return its initial investment from its cash flows. (*Chapter VIII*)

Perpetuity. A series of equal payments that continues for ever. (*Chapter XIII*)

Portfolio. A collection of investments held together by one investor. (*Chapter XI*)

Preferred stock. Shares that pay a fixed dividend, which must be paid before any dividend on common stock. (*Chapter X*)

Premium bond. A bond that sells for more than its face value. (*Chapter XII*)

Present value. What a sum due in the future is worth today. (*Chapter V*)

Principal. A person who hires another to act for them. A shop owner who hires a manager is a principal. (*Chapter III*)

Principal sum. The original amount lent, borrowed or deposited, before any interest is added. (*Chapter IV*)

Probability distribution. A list of possible results with the chance of each, the chances adding to one. (*Chapter IX*)

Profit. What a business earns from sales after subtracting the costs of earning them. Sales of Tk 100 and costs of Tk 80 leave profit of Tk 20. (*Chapter I*)

Profit maximisation. Aiming to make the largest profit possible, usually within one year. (*Chapter III*)

Profitability index. Present value of future cash flows divided by the initial investment. (*Chapter VIII*)

Rate of return. The return in one period divided by the sum invested, written as a percentage. (*Chapter IX*)

Relevant cash flow. A cash flow that changes because of the decision being studied. (*Chapter VII*)

Required return. The least return an investor will accept for taking a given risk. (*Chapter XI*)

Residual claim. The right to whatever is left after lenders and preferred owners have been paid. (*Chapter XIII*)

Retained earnings. Profit kept in the business instead of being paid out as dividends. (*Chapter XV*)

Return. The gain or loss from an investment, including income received and any change in value. (*Chapter IX*)

Rights issue. An offer of new shares to existing shareholders, in proportion to their holdings. (*Chapter XV*)

Risk. The chance that the actual result will differ from the result you expect. (*Chapter IX*)

D

Debenture, 119

Depreciation, 52, 56, 128, 131

Discount bond, 94

Discount rate, 35, 40, 59, 90

Discounted payback period, 58

Discounting, 24, 34–36, 39–40

Diversification, 85–86

Dividend, 10–11, 79, 101–102, 119

Dividend decision, 11

Dividend valuation model, 101–102

E

Earnings per share, 79

EBIT, 77–78

Efficient market, 87

Equity, 12, 90, 105, 119

Expected value, 68, 72

F

Face value, 93–94

Factor table, 29, 32

Finance, 2–4, 7–8, 11–13, 15, 19–21, 23–24, 27–29, 32, 35–37, 40, ...

Financial lease, 127, 131

Financial leverage, 78

Financial manager, 10–13, 15

Financial planning, 11

Financial risk, 67, 72

Financing decision, 10

Fixed cost, 76–77

Float, 110–111, 114

Future value, 4, 13, 21, 26, 28–29, 31–32, 36, 43, 45, 48, ...

I

Incentive, 20

Independent project, 50

Initial investment, 51, 58–60

Initial public offering, 120

Instalment, 44, 48, 118

Interest, 19, 26–29, 31–32, 34, 40, 67, 77, 93–94, 98, 128, ...

Interest rate, 27

Intermediate-term financing, 109

Internal rate of return, 59, 64

Underwriter. An institution that promises to buy any shares of a new issue that the public does not. (*Chapter XV*)

Unsystematic risk. Risk that affects one firm or industry only; diversification can remove it. (*Chapter XI*)

Valuation. Finding what an asset is worth today by adding the present values of the cash it will bring. (*Chapter XII*)

Variable cost. A cost that rises and falls with the number of units sold, such as the price paid for each book. (*Chapter X*)

Variance. The probability-weighted average of the squared gaps between each result and the expected value. (*Chapter IX*)

Wealth maximisation. Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is. (*Chapter III*)

Weighted average cost of capital. The average cost of a firm's different sources of money, each weighted by its share of the total. (*Chapter XV*)

Working capital. The money a business has tied up in everyday trading: cash, stock and money owed to it. (*Chapter XIV*)

Yield to maturity. The yearly return an investor earns by buying a bond at today's price and holding it to maturity. (*Chapter XII*)

ANSWERS TO QUESTIONS

CHAPTER I · WHAT FINANCE IS

01.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

01.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

01.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

01.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

01.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER II · THE FINANCIAL MANAGER'S DECISIONS

02.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

02.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

02.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

02.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

02.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER III · THE GOAL OF THE FIRM AND AGENCY THEORY

03.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

03.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

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13.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

13.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

13.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

13.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XIV · WORKING CAPITAL AND SHORT-TERM FINANCING

14.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

14.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

14.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

14.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

14.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

14.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XV · LONG-TERM FINANCING AND THE CAPITAL MARKET

15.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

15.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

15.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

reasoning should use the facts given in the question and keep timing and units consistent.

06.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

06.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

06.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

06.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

06.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER VII · CAPITAL BUDGETING: PROCESS AND CASH FLOWS

07.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

07.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

07.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

07.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

07.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

07.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER VIII · EVALUATING PROJECTS: PAYBACK AND NET PRESENT VALUE

08.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

08.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

reasoning should use the facts given in the question and keep timing and units consistent.

08.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

08.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

08.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER IX · RISK AND RETURN

09.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

09.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

09.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

09.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

09.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER X · LEVERAGE

10.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

10.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

10.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

10.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

10.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.